Fiscal Year:	FY 2004	**Task Last Updated:**	FY 04/03/2006
PI Name:	Schlegel, Robert E. Ph.D.		
Project Title:	Integrated Crew Performance Assessment and Training		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Space human factors engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	73019	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-01
Start Date:	01/01/2001	**End Date:**	12/31/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Shehab, Randa (University of Oklahoma) Gilliland, Kirby (University of Oklahoma)		
Grant/Contract No.:	NNAG 9 1280		
Performance Goal No.:			
Performance Goal Text:			
	As future space missions become longer and more complex, the human performance requirements associated with those missions are changing. Detailed training for all mission tasks is not always completed pre-flight, but in many cases is delivered "just-in-time" (JIT). The changes in mission requirements, when coupled with the changes in operator state associated with longer missions, underscore the importance of an accurate astronaut status assessment system. An integrated system that presents recommendations for new or refresher training for upcoming activities would provide even greater utility. The overall objectives of the current research are to further the self-assessment of astronaut cognitive and sensorimotor state as it relates to crew performance, and to establish a framework for integrating the assessment results with prescriptions for training and countermeasure interventions. In the arena of self-assessment, it is important for each astronaut to determine his or her current state in relation to individual capacity. This necessitates an assessment of		

Task Description:	reserve capacity in a relevant mission setting. Traditionally, reserve capacity has been assessed using a secondary task method that assesses status based on performance on an additional, non-primary task. In contrast, the proposed solution in this research is the development of "dynamic load" tasks, whose level of difficulty changes as the task progresses. The research approach involves innovative transformation of traditional cognitive assessment tasks in order to measure critical limits at the time of testing in relation to an unstressed baseline obtained at the maximum trained proficiency. The initial focus of our research is on the self-assessment of sensorimotor or visuomotor state. We will incorporate validation of the instrument using a training simulator for RMS (robot arm) operations. The research hypothesis is that appropriately constructed dynamic-load tasks will reveal cognitive decrements sooner than simple tasks, that is, at an earlier point of training proficiency loss or at a lower exposure to a risk factor. This hypothesis will be tested by comparing the sensitivity of dynamic-load tasks with the sensitivity of standard static-load tasks utilizing similar task elements. For validation purposes, the sources of performance change will be time-induced skill decay and fatigue-induced changes in visuomotor and cognitive state. In summary, the research will provide the assessment methodology and implementation for one dimension of cognitive/sensorimotor performance (Year 1), implementation of the methodology with a simulated NASA operational task (Year 2), a ground-based validation study (Year 2), and a multivariate expansion of the system (Year 3).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our current NASA research forms a component of a larger program in cognitive performance assessment at the University of Oklahoma. Successful development of this next generation of dynamic-load assessment tasks will lead to expansion of the methodology to other dimensions. Final implementation in an operational setting will require integration of the task modules into an automated planning, scheduling, and resource allocation system with an intelligent interface to improve astronaut autonomy while maintaining high levels of mission success. This research will also lead to improved cognitive performance assessment in military and commercial contexts. Results of the research will be readily usable in readiness-to-perform (RTP) programs in the public sector. Demonstration of the effectiveness of complex, dynamic-load tasks and other strategies for assessing performance changes will translate directly to RTP tests with greater sensitivity and selectivity, especially with the development of multivariate scoring approaches.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/03/2020)
Articles in Peer-reviewed Journals	Bray TM, Shehab RL, Schlegel RE, Civan A, Walker DE. "Effectiveness of 2-D views for 6-D robotics simulation maneuvers." Proceedings of the Human Factors and Ergonomics Society Annual Meeting. 2003 Oct;47(8):975-9. (47th Annual Meeting of the Human Factors and Ergonomics Society, Denver, Colorado, October 13 - 17, 2003.) https://doi.org/10.1177/154193120304700805 , Oct-2003